

Ishan Trivedi

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Hugging Face](#) | [X](#)

EDUCATION

Sir M. Visvesvaraya Institute of Technology, Bengaluru

2024 - Present

B.E. in Computer Science & Engineering, Visvesvaraya Technological University - CGPA: 8.5/10

Relevant Coursework: Data Structures, Linear Algebra, Discrete Mathematics, Object-Oriented Programming

PROJECTS

IncidentOps (Ongoing) | Python, FastAPI, PostgreSQL, Qdrant, Redis, LangGraph, Azure

[Core](#) | [Collector](#) | [Live](#)

- Building a V2 engineering-evidence backend that ingests code, logs, deploy records, runbooks and incident context through a deterministic Collector, then returns cited search and investigation results with explicit missing-evidence warnings.
- Designed hybrid retrieval across **5 evidence signals** (vector, lexical, metadata, path and graph) with routing across **9 query-intent classes**; deployed an Azure proof path with **4 application services** (Core API, worker, private MCP and Collector) and **2048-dimensional Nemotron embeddings**.

Agent Canary | Python, LangGraph, FastAPI, NVIDIA Nemotron, React

[GitHub](#) | [Explainer](#) | [Dashboard](#)

- Built an authorized AI-agent red-team platform where a **4-role LangGraph workflow** generates adversarial prompts, sends them to real HTTP targets, evaluates actual responses and persists evidence-backed findings for human review.
- Implemented **12 registered attack strategies**, **8 dashboard-selectable techniques** and up to **12 parallel attacker branches**; combined deterministic detectors with a Nemotron evaluator while recording target response, HTTP status, latency and verdict evidence.

LogSage | Python, PyTorch, QLoRA, Hugging Face, AWS EC2

[GitHub](#) | [Explainer](#) | [HF](#)

- Fine-tuned **Qwen2.5-7B-Instruct** with 4-bit QLoRA (r=16) for structured log triage, producing schema-validated JSON with issue, root cause, severity, fix and confidence instead of raw text.
- Curated **1,116 examples** (1,004 train / 112 eval) and trained for **378 steps / 3 epochs** on an AWS A10G, reaching **0.789 best eval loss in 34.3 minutes**; published the adapter and reproducible inference workflow to Hugging Face.

AudioForge | Python, PyTorch, LoRA, Hugging Face, FSD50K

[GitHub](#) | [Explainer](#) | [HF-CNN](#) | [HF-AST](#)

- Built a reproducible training framework for **200-class FSD50K**, comparing a **2.4M-parameter** scratch CNN with an approximately **86M-parameter** Audio Spectrogram Transformer adapted with LoRA.
- Improved held-out mAP from **0.3020 to 0.5567 (84.3% relative gain)** with the LoRA-adapted AST; implemented data prep, mixed-precision training, checkpoint resume, validation, reports and Hugging Face publishing.

TECHNICAL SKILLS

LLM / Retrieval / Agents: LangGraph, RAG, hybrid retrieval, Qdrant, pgvector, embeddings, tool calling, LLM evaluation, structured outputs, MCP

AI / ML: PyTorch, Hugging Face Transformers, PEFT, LoRA/QLoRA, supervised fine-tuning, CNNs, Audio Spectrogram Transformers, model evaluation

Backend / Infra: FastAPI, PostgreSQL, Redis, REST APIs, JWT/RBAC, background workers, Docker/Compose, Azure Container Apps, AWS EC2, NVIDIA NIM, Vercel, Git/GitHub

Programming: Python, Java, TypeScript, SQL

DSA: Arrays, Strings, Hashing, Two Pointers, Sliding Window, Binary Search, Linked Lists, Stacks & Queues, Trees, Heaps, Graphs, Recursion, Backtracking, Greedy, Dynamic Programming | [LeetCode](#)

Data: NumPy, pandas, scikit-learn, JSON/JSONL, Jupyter/Colab